

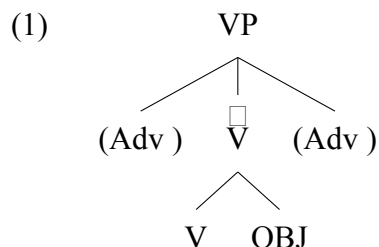
Head movement and linear edges

September 30, 2025

1 Introduction: Head Movement

1.1 Motivating Head movement

It is a well-known fact that verbal heads do not necessarily appear in the position where they are introduced in the tree. Given that verbs select objects in their complements, and that adverbs like *often* are located at the VP/vP-edge (1), verbs would be expected to appear in a position strictly adjacent to the objects. That is indeed the case for English lexical verbs, as shown in (2).



- (2)
- a. Mary often reads books
 - b. *Mary reads often books
 - c. Mary reads books often

However, in many other languages, comparable adverbs appear between verbs and objects. This is the case in French, for instance, where the adverb may appear between the verb and object (Emonds 1970; Pollock 1989). In fact, the adverb cannot appear to the immediate left of the verb:

- (3)
- a. Jean embrasse souvent Marie
J. kisses often M.
 - b. *Jean souvent embrasse Marie
J. often kisses M.
 - c. Jean embrasse Marie souvent
J. kisses M. often
"John often kisses Mary"

Hence, the linear position of the verb doesn't reflect constituency.

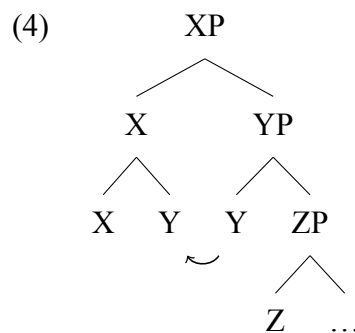
The general solution to such apparent linear order - constituency mismatches is movement (Internal Merge), where in our cases, the head adjoins to the next higher head (Koopman 1984; Travis 1984; Baker 1985). This type of account is known as *head movement*. In the French examples, V adjoins to a higher head (say T) and, as a result the verb appears to the left of a VP-level adjunct.

1.2 Challenges for Head Movement accounts

Traditionally, head movement has been perceived as a movement operation applying in narrow syntax, much akin to A- or $\bar{A}$ -movement. At the same time, there are at least four dimensions in which head movement deviates from other types of movement.

1.2.1 Cyclicity

The first one is that head movement violates common conceptions of cyclicity — we dub this the *Cyclicity Problem*. As can be seen in the tree below, head adjunction does not target the root of the tree and in that sense, head movement is not cyclic: it violates the extension condition.



1.2.2 Semantic vacuity

The second dimension is that unlike cases of A- or $\bar{A}$ movement, head movement does not appear to feed LF. Head movement appears semantically vacuous — we term this the *Vacuity Problem*. This can be seen in examples like (5)-(6):

(5) Mary may not eat any cookies

In (5), the modal *may*, an existential quantifier over possible worlds, underwent head movement across negation, but it does not allow a reading where it takes scope above negation. This contrasts with cases of phrasal movement of an existential quantifier over negation, which does lead to surface scope readings:

(6) A student has not passed the exam

The claim that head movement does not feed semantics is not uncontroversial, though. For instance, it has been argued in the literature that head movement can sometimes feed semantics.

Lechner (2004) and Iatridou and Zeijlstra (2013) for instance argue that scopal relations between negation and modals are best captured under the assumption that head movement feeds semantics (though see for a criticism of such claims).

For instance, Iatridou and Zeijlstra (2013) argue that *must* in (7a) moves from a position below negation to one above it. Being a Positive Polarity Item (PPI), it cannot reconstruct below negation and takes scope in situ, unlike polarity-insensitive modals like *may* and *can*, which must reconstruct below negation.

- (7) a. Mary *mustn't* leave
b. Mary *can't* leave

However, those arguments are not too strong, as already pointed out by Iatridou and Zeijlstra (2013, fn. 22), since it can also be maintained that *must* undergoes postsyntactic QR irrespective of where it appears in its surface position, much in the same vein as *some girl* in *I didn't see some girl* can QR across negation (see Homer 2015). In fact, Jeretič and Thoms (2023) show that an analysis in terms of head movement feeding semantics actually suffers from various problems.

This shows that any argument that is based on head movement of quantifying elements can be subject to an alternative analyses where the relevant element undergoes overt but semantically vacuous head movement and covert QR. Also in other cases, alternative analyses can apply equally well, if not better, for instance in cases where moving a head to a higher position affects the way elements take scope.

Matyiku (2016) presents data from Western Texas English where subject-auxiliary inversion disambiguates particular readings, for instance in the case of universal subjects in negative clauses:

- (8) a. Everybody *didn't* leave $\neg > \forall; \forall > \neg$
b. *Didn't* everybody leave $\neg > \forall; *\forall > \neg$

Under Matyiku's account, head movement of *didn't* across the subject position, renders an inverse scope reading of negation and the quantifier impossible. However, if (8b) is derived from (8a) and (8a) allows both scopal positions, (8b) should in principle still allow those as well. In both cases there is a derivational stage where the subject *everybody* c-commands the negative head *n't*. The absence of the inverse scope reading in (8b) can thus not be the sole effect of head movement.

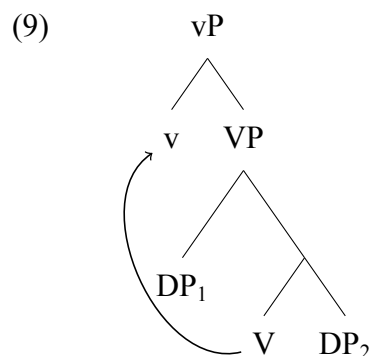
The reason why the inverse scope reading of (8a) is not available, must then also have to do with the fact that the (pragmatic) mechanism that determines which of the readings that can be grammatically derived end up being available in a language, must make reference to surface orders (see Fanselow and Cavar 2001, 2002; Bobaljik and Wurmbrand 2012 a.o.). Without such a mechanism any clausal combination of a quantifier and another scope-taking element in any language, where one moves across the other, should be ambiguous, contrary to fact. That means that even though indirectly, head movement may have semantic effects in that it blocks certain readings, this is not a direct semantic effect. Hence, head movement cannot be said to feed semantics in these cases in the sense that it plays a role at LF. This also relates to cases where a Negative Polarity Item (NPI) subjects can appear under negated auxiliaries, as in *Which one of them doesn't anybody like?* (McCloskey 1996, see also Roberts 2010). It is independently known since Ladusaw 1979 that NPI licensing is subject to surface orders (his so-called surface roof constraint), even though

the mechanism responsible for the anomaly of unlicensed NPIs applies at LF (see Chierchia 2013 a.m.o.)¹

The same kind of reasoning also applies to the cases discussed in Szabolcsi (2011) of aspectual raising verbs like *begin*, modal auxiliaries and potentially also intensional raising verbs where the position of the verb determines whether they are scopally ambiguous with respect to the subject or not. Also, Keine and Bhatt (2016) show in German long passive constructions that when the lower verb undergoes head movement into a matrix verb, arguments and modifiers of this lower verb must also take scope above this matrix verb. Again, here head movement does not (necessarily) feed semantics. Rather, the scopal properties of other elements depends on the surface position of the verb. In this sense, these data do not provide evidence that head movement itself must be semantically active.²

1.2.3 Phrasing

Another challenge for head movement, our third dimension, is noted by Kalivoda (2018). He notes that, given a Match Theoretic view of the mapping from syntax to prosody (Selkirk 2011), standard approaches to ditransitive constructions, which are commonly assumed to involve head movement low in the clause, are predicted to have a particular prosodic structure, where both internal arguments should form a joint prosodic phrase — this we name the *Phrasing Problem*.



As Kalivoda notes, two of the most commonly assumed models of deriving prosodic constituents from syntactic constituents — namely *Align* (Truckenbrodt 1995) or *Match* (Selkirk 2011)

1. Note however, that this is not the only possible explanation for the facts mentioned above. Following Penka (2011), Zeijlstra (2022b) has argued that English *n't* is not the realization of a semantic negative operator but rather a semantically non-negative marker of its presence. That is, a semantically negative, covert operator is base-generated above the surface position of *n't*. If that is the case, it follows trivially that in (8a) the subject must take scope below negation, but in (8b) it doesn't. The licensing of NPI subjects would follow in the same way.

2. These are not the only claims that head movement feeds semantics. More indirect claims have been made as well, for instance by Gribanova 2017, who argues, following Hartman 2011, that head movement may affect the way constituents can meet the semantic identity requirements that license ellipsis, but also here the effect is indirect and there are other ways to derive the relevant facts without alluding to LF reflexes of head movement. Other indirect relations between head movement and LF effects consist of order preservation effects (like Holmberg's Generalization), where phrasal movement appears dependent on head movement. If such phrasal movement can feed semantics, head movement should apply in Narrow Syntax as well. At the same time, such effects can also be understood in terms of order preservation at PF, where the surface position of a head must remain linearly fixed with respect to the surface position of some phrasal element along the lines of Fox and Pesetsky 2005 (see also Dikken 2007 for more discussion).

— make undesirable predictions about the typology of prosodic phrasing of ditransitives when head movement of the sort schematized above is assumed. *Match* makes a pernicious prediction that the following prosodic phrasing for ditransitives should be possible and perhaps widely attested, yet, as Kalivoda shows in a comprehensive survey of such patterns, no language is known to exhibit such phrasing. For the tree above, after head movement has taken place, there is a syntactic constituent containing only the two internal arguments, to the exclusion of the verb; *Match* theory would lead us to expect this constituent to map to a prosodic phrase, yet no such phrasing is known to exist.

$$(10) \quad (V (DP_1 DP_2))$$

In contrast to the overgeneration problem that befalls *Match* theory, Kalivoda notes that *Align*-based approaches undergenerate in the case of patterns like those found in Ewe ditransitives, schematized below. The problem for such an approach is that there is no syntactic edge between the verb and DP_1 that should map to the sort of prosodic boundary Ewe tone sandhi seems to be sensitive to.

$$(11) \quad (V) (DP_1) (DP_2)$$

These problems aside, Kalivoda notes that the most common patterns for prosodic phrasing of ditransitives are those shown below, where the verb either phrases together with the linearly closest argument, or with both internal arguments. He notes that accounting for the commonality of these patterns on an *Align* or *Match* based theory require auxiliary constraints, which have most often been cashed out as involving to syntactic relationships such as selection, and suggests that such constraints are undesirable for reasons related to modularity.

$$(12) \quad (\varphi \ v+V \ DP_1 \ DP_2)$$

$$(13) \quad (\varphi \ v+V \ DP_1) (\varphi \ DP_2)$$

Kalivoda takes these facts to motivate a different approach to the mapping of syntax to prosody, which we take no particular stance on here. What we wish to note is that the three mappings above — those in (12) as well as the Ewe pattern in (13) — are exactly what we might expect were head movement not to take place at all, supplemented by other assumptions about deriving prosodic constituency from syntactic constituency common to Kalivoda's model as well as *Match* and *Align*-based models. Moreover, the phrasing found in (10) under a *Match* theoretic approach would be unexpected given a model where head movement not to take place. This, foreshadowing the proposal to be introduced later in this paper, would be possible only if heads could be linearized to the left of their specifiers.

1.2.4 Locality

The fourth dimension in which head movement is different from A- or $\bar{A}$ movement concerns locality. Head movement is strictly local. It is impossible to adjoin a head X to a head Y if some head Z intervenes. This is known as the Head Movement Constraint (HMC), which goes back to Travis (1984). A- or $\bar{A}$ movement, by contrast, does not have to target the closest specifier (albeit A-movement does target the closest A position). We call this the *Locality Problem*.

This difference between head movement on the one hand and A- and A'-movement on the other have led to the development of three schools of thought. One places head movement squarely in

the syntax, with the differences between head movement and phrasal movement being attributed to other factors; the other is to delegate parts of head movement to a distinct module of the grammar. A third approach takes head movement — at least of the classical sort — out of narrow syntax in its entirety. Below we discuss these briefly, and relate them to the proposal we wish to make.

1.3 Approaches to head movement and their problems

Here, we'll briefly review several schools of thought on head movement and how they relate to the problems mentioned beforehand. For a more detailed overview, we refer the reader to Dékány (2018).

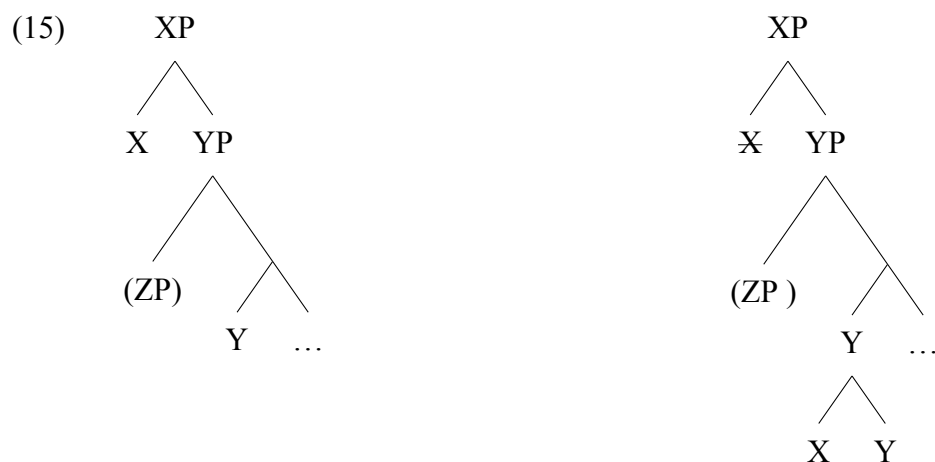
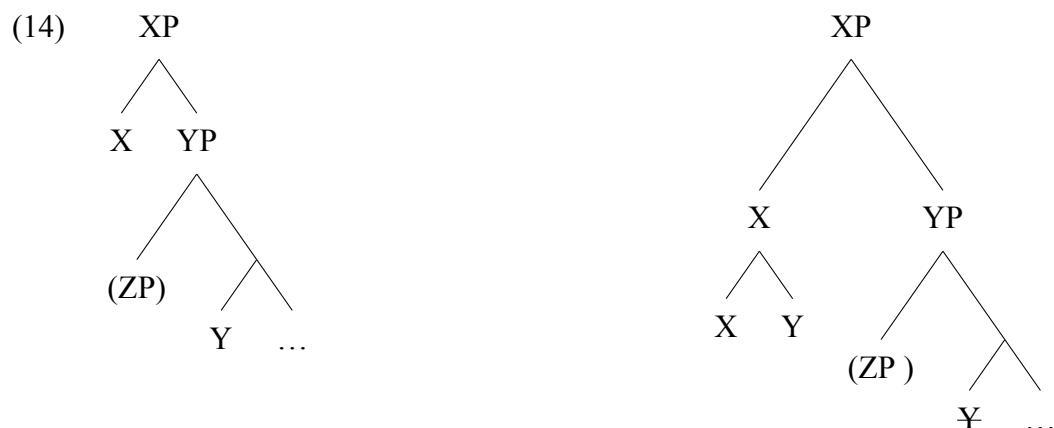
One sort of approach locates head movement chiefly in the syntax. This could be done by adopting an approach where Merge is allowed to apply freely (see Safir (2019) for an instance where this logic is made explicit). Another mechanism to resolve the *Cyclicity* problem has been proposed in terms of so-called the reprojection approach to head movement (Koenenman 2000, Fanselow 2004, Suranyi 2008). Under such an approach, a verb does not adjoin to a higher head, but rather adjoins to the phrase it is the head of and projects again. Crucial under this approach is that heads enter the derivation with all the relevant features (e.g., [V], [v], [T]). Every time a different feature then projects, thus yielding a TP dominating a vP, which in turn dominates a VP, albeit that it is not clear why such a head should first project a VP, then a vP and then a TP. As for all these approaches, while it is possible to define a cyclic principle regulating Merge that will allow for both head movement and phrasal movement, it is not clear that such an approach escapes the other problems for syntactic head movement mentioned beforehand, most notably the *Vacuity Problem* and the *Phrasing Problem*.

Another way to circumvent the *Cyclicity* problem within narrow syntax would be to assimilate head movement to phrasal movement (see Massam (2000) for such an analysis of a verb-initial language). While we do not dismiss such an approach for languages like Niuean and comparable verb-initial languages, we suspect that a full-blown extension of such an account for all instances of head movement ultimately be unsatisfactory. Another approach along these lines would be Roberts (2010), who argues that head movement reflects the same underlying Agree mechanism as phrasal movement. For him, when the features of a goal are copied on the probe, these features can also be spelled out on the problem. In that case, he argues, the goal incorporates into the probe. However, this incorporation may still suffer from the *Cyclicity Problem* and the *Locality Problem* as incorporation violates the Extension Condition and probe-goal relations do not have to apply in a strictly local fashion (although Roberts argues that Extension Condition and the Head movement Constraint do not always apply). Since Agree can feed semantics, it remains unclear as well under this proposal how the *Vacuity Problem* is resolved, as well as the *Phrasing Problem* if the goal really incorporates in the probe.

A second sort of approach places portions of head movement in the syntax and other portions in the morphology. The most famous example of this sort of approach comes from Matushansky (2006). Matushansky suggests that head movement takes place in two steps. Heads may undergo movement to specifier positions, with a subsequent morphonological operation — M-merger — being responsible for assimilating the head in specifier position together with another nearby head. While this approach solves the *Cyclicity Problem*, it runs afoul of the three other problems mentioned before. It does not lead us to expect head movement to be semantically vacuous, so it is still subject to the *Vacuity Problem*, it predicts the unattested prosodic phrasings that characterize the

Phrasing Problem, and the *Locality Problem* is not addressed under this proposal either.

A third sort of approach locates head movement in the postsyntax. Thereby, the differences between head movement and non-head movement are captured by attributing the properties of one to a distinct module from the other. Outsourcing head movement to PF trivially explains why it no longer feeds LF, resolving the *Vacuity Problem*. It may also avoid the *Cyclicity Problem* as there is no reason to a priori expect operations at PF to be subject to the same cyclic principles as the narrow syntax, although it is not clear why cyclicity should work differently in the PF component of grammar. The *Phrasing Problem* might also be worked around if the prosodic structure is derived from a PF representation, rather than the syntactic tree proper (although we do not know of any claims made in the literature which directly address this juncture between PF-level rebracketing and prosodic phrasing). This third sort of approach thus seems more promising, as it furnishes a straightforward solution to the three problems discussed beforehand. Indeed, we will develop just such an approach. However, we wish here to take issue with many approaches to post-syntactic head movement, namely that the structures which they operate on are similar in nature. Taking Harizanov and Gribanova (2019) as an example, consider the operation they propose for instances of head movement at PF: given two heads, where the first is the head of the phrase which serves as the complement to the second, one of the two might be adjoined to the other.

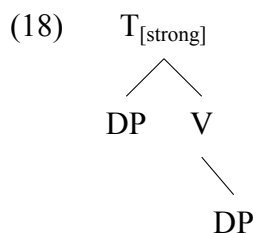
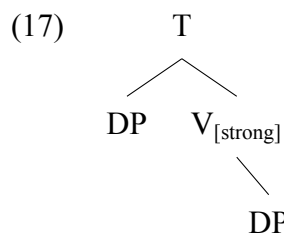


There are good reasons to be suspicious of approaches such as these, however. It is not clear that much is to be won by just taking a set of putatively syntactic operations and declaring them to be part of a distinct module. We are therefore suspicious of putatively modular approaches to grammar where two supposed modules of grammar truck in the same vocabulary — here, minimal and maximal projections, and adjunction to them — and produce very similar outputs — here, novel constituents where the properties of one of the two daughters of the novel constituent are inherited.

A fourth sort of approach pioneered by Brody (2000) involves a revision of the syntactic apparatus. Doing so allows for a linearization algorithm which may produce head movement effects as a byproduct of linearization. Crucial assumptions for this type of approach are enumerated below.

- (16) a. A node may represent both a head and its phrasal projection.
- b. Phrasal specifiers precede the head they are a specifier of, phrasal complements follow heads they are a complement of; head complements precede heads they are a complement of, head specifiers follow heads they are a specifier of.
- c. Language specific parameters may designate heads in an iterated complement relationship as morphological words.
- d. Heads may be marked as “weak” or “strong”, and this determines where a morphological word is linearized with respect to phrasal elements.

Below we provide representations for an English-like language and a French-like language. Their syntax is identical, with one DP serving as the specifier of T (indicated as a left branch on the tree), and another serving as the complement of V (indicated as a right branch on the tree). Both are assumed to be languages where T and V form a morphological word. The chief difference between the two is whether T or V is marked as strong. In an English-like language, (17) the exponents of T and V will precede the complement of V, but not the specifier of V, while in a French-like language, (18), the exponents of T and V will precede the complement of T (namely, both the specifier and complement of V).



Additionally, such approaches may run afoul of the *Phrasing Problem*, depending on how the mapping from syntax to prosody is formulated for Mirror Theoretic representations. On a straightforward approach where “strong” in (17d) is taken to indicate that such a position is privileged for the PF branch, either the overgeneration problem inherent to Match-theoretic approaches should arise, or the undergeneration problem inherent to Align-theoretic approaches should arise.³

Our aim, then, is to sketch what a post-syntactic approach to head movement might look like without making a direct appeal to operations of the sort discussed above. We will take an altogether different approach, and place much of the responsibility for ensuring that two heads are in a local

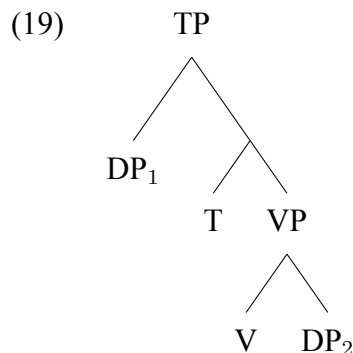
3. Moreover, it is unclear whether a Command-theoretic approach like that developed by Kalivoda could be straightforwardly recast to be compatible with Mirror-theoretic syntactic objects.

relationship in the hands of a different PF mechanism without alluding to any kind of movement, either in narrow syntax, or at PF. Instead, we suggest that head movement effects might be treated as the consequence of a particular linearization algorithm that allows a wider range of possible linear orders for a given maximal projection than is commonly assumed.

The theory developed here resembles that proposed by Brody and others in developing a system where some head movement effects result from a theory of linearization. We will see that ours is distinct from theirs in a number of ways. The theory to be developed makes certain predictions not only about where heads may or may not be pronounced with respect to other heads, but also bars phrases from appearing in certain positions in a way that a mirror-theoretic approach does not — allowing us to understand, for instance, differences in the possible interpretations of scrambled sentences in different languages, discussed in §4.

2 Linearization and head movement

Following Kayne (1994) and much subsequent work, it is often assumed that there is a strict one-to-one mapping of syntactic structure to linear order. For instance, if one has a tree like the following, only one order of terminal nodes will be admitted, namely: DP_1 T V DP_2 . Under such strict views of linearization, *any* deviation from this linear order must reflect a difference in syntactic structure — head-finality, for instance, must involve movement of constituents within the clause.



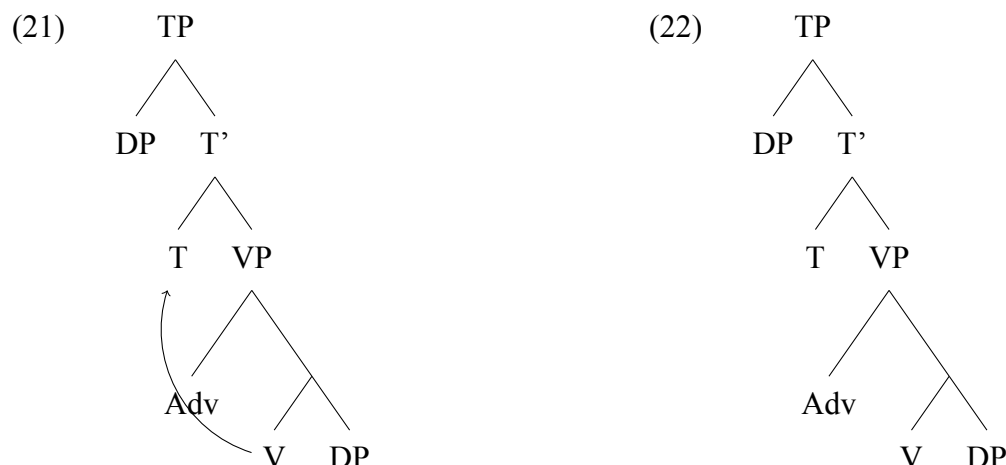
Even if a less strict approach is taken — for instance, by assuming something like a head parameter — the result is the same: any one tree must map to exactly one linear order. Linearization requires an algorithm, either as a strict mapping or by means of a parameter. But if we admit a head parameter into the system that is not uniform across all heads in a language, in essence allowing the features of certain lexical items to determine linear order, we might wonder whether a fully feature-driven view of linearization is possible. We will pursue this line of analysis here. More specifically, we will suggest that the relative order of heads and phrases dominated by the projection of that head are, at least in principle, freely linearizable. Restrictions on the ordering of elements arise from lexically specified features analogous to the aforementioned features encoding the head parameter, and a highly cyclic approach to the linearization of elements in the clause. Our core focus, naturally, will be capturing of certain word order patterns generally attributed to head movement.

2.1 Linearization

As discussed before, a well known set of facts commonly attributed to head movement involves differences between languages with respect to possible orderings of the head, complement, and adjunct. For expository purposes, we restrict ourselves to fairly simple syntactic representations free of intermediate projections, but we note that the analysis presented here is, of course, compatible with enriched representations of this sort. As shown in Section 1, and repeated here, certain adverbs may appear between the verb and object in French, but not in English (at least when it concerns lexical verbs).

- (20) a. Jean embrasse souvent Marie
 J. kisses often M.
 “John often kisses Mary.” French
 b. *John kisses often Mary.

A common analysis of facts like these, following Emonds (1970) and Pollock (1989), involves a parametric difference between English and French cashed out in terms of head movement: in French, V moves to T (21), while in English V does not (22).



The strategy here, then, is similar to that commonly assumed for A and $\bar{A}$ -movement. The expected linear order where the adverb precedes the verb is not present in French as a result of the verb undergoing displacement to a head which takes the VP, including the adverb, as its complement. The relevant linear order reflects derived constituency.

But as said before, the question is indeed whether linearization always needs to reflect (derived) constituency. Going back to the head parameter, the idea is that linear order is not fully predictable on the basis of the hierarchical order of the syntactic objects, but only partially so, and additional linearization constructions come from features on particular heads. If V has some head-initial feature, it must be linearized to the left of its complement (i.e. to the left of the material that makes up the complement); if V instead has a head-final feature, it must be linearized to the right of it.

Head-initial and head-final features are, however, not sufficient to determine the linear order between a head and its specifier(s) and/or adjuncts. Unless additional mechanisms are assumed

(like anti-symmetry or other linearization features) the order of a head with respect to these non-complements (both specifiers and adjuncts) is underdetermined. The question is whether that is a good or a bad thing.

But before addressing that question, it should be noted that specifiers and adjuncts behave differently when it comes to (rigid) linearization. Specifiers generally have a specific position in the linear order (often to the left of their heads); here underspecification would be problematic:

- (23) a. She likes her
b. *Likes her she

But adjuncts are relatively free and can often appear both to the left and to the right of the verb and its complement, as shown below:

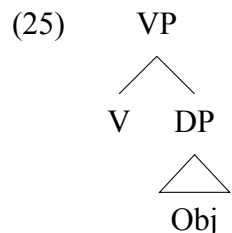
- (24) a. She often eats spinach
b. She eats spinach often

From the perspective of adjuncts, underspecified linearization constraints would actually result in the observed word order freedom.

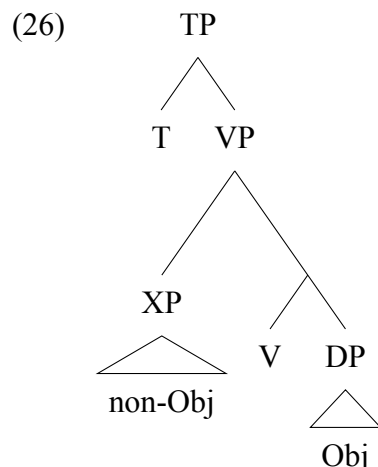
There are good reasons to assume, though, that the fixed order of specifiers is due to other factors. For one, specifiers could tend to be leftward for reasons like the EPP. Alternatively, certain affixes need something metrically strong supporting them or a prosodic alignment requirement (Richards 2016). Another reason for the leftness of specifiers in head-initial phrases could be that certain heads are barred from appearing phrase initially (see Sabbagh 2014, who discusses 2nd person clitics of various sorts). And finally, as argued for in Zeijlstra 2022a, if specifier positions are landing sites for movement, and movement is leftward, such specifier positions should be to the left of their complements as well. Irrespective of the exact reason, we take the general leftness of specifiers to be something that follows on independent grounds and argue that a general theory of linearization can leave the position of non-complements in principle relatively free. In §4, we make an explicit proposal for why derived specifiers tend not to appear on the right, grounded in the feature-based theory of linearization we put forth here.

The question, then, arises how the linear order of heads, complements and non-complements (i.e. specifiers and adjuncts) is determined. To answer that question, we will discuss our assumptions step by step and look first at cases of English lexical verbs, where no head movement effects can be observed. After that, we will look at cases where head movement effects do play a role.

Our first assumption is that linearization takes place whenever a head and complement are merged. The reason is that when a head and its complement merge, it is only then that the head parameter can be properly implemented. As we will see, the fact that linearization is highly cyclic also makes what looks like head movement highly local. Hence, when a VP consisting of a head V and a complement DP is linearized, V is either linearized to the left of (all the material inside) DP or to the right of it. In English, V, being head-initial, ends up at the left of its complement.



The next step of linearization applies when the next head is merged, which we take to be T:



It is already clear from the above that V must appear to either the left or the right of its complement. Since English is overall head-initial, V appears to the left. The same applies to T. This way, the following linearization statements are established:

- (27)
- a. $T < VP$
 - b. $V < DP$

As VP contains XP, V and DP, the list of established linearization conditions can be extended as follows:

- (28)
- a. $T < XP$
 - b. $T < V$
 - c. $T < DP$
 - d. $V < DP$

Hence, the order $T < V < DP$ is already established; the only question is where XP needs to be pronounced. In this light, it is important that the relative position of XP does not have to be restricted to one position. As we saw, in the case of adjuncts, it can be either in between T and V, or after the DP. In other words, the adjunct needs to be linearized at the edge of the VP. The word *edge* is deliberately chosen.

But what is the proper definition of “edge”? It may be intuitive to think of the edge of a phrase as the outer layer (either linear or structural) of some phrase. But outer is configurationally defined;

a position is an outer position of a phrase if the next element up no longer belongs to that phrase. Let's therefore define a linear edge as follows.

- (29) An element X that is part of a phrase BP in a configuration [A [BP]] is at the edge of BP if it is
- a. linearized as closely to A as possible with respect to other linearization conditions,
 - b. or as far away as possible from A with respect to other linearization conditions.

This way, XP is either linearized to the immediate right of T, or to the immediate right of the DP. This gives us two sets of possible linearization conditions:

- (30)
- a. $T < XP$
 - b. $T < V$
 - c. $T < DP$
 - d. $V < DP$
 - e. $XP < V$
 - f. $XP < DP$

or

- (31)
- a. $T < XP$
 - b. $T < V$
 - c. $T < DP$
 - d. $V < DP$
 - e. $V < XP$
 - f. $DP < XP$

This is indeed in full accordance with the examples shown before. If XP is an adjunct, both orders are fine; if XP is a specifier, the second order must be ruled out independently, for reasons we have mentioned earlier in this section. This allows us to formulate the following linearization algorithm:⁴

- (32) **Within a cycle of linearization**
- a. A head is linearized to the left or right of its complement.
 - b. A non-complement inside the complement of a head is linearized at an edge of that complement.

This derives the English cases with lexical verbs that do not display head movement effects.

In cases where there is more than one specifier/adjunct, our system predicts that the highest one will always be linearized either adjacent to the head (modulo head movement effects, see later) or as far away from it. This prediction is borne out given the existence of certain mirror effects

4. Note that to the extent that one wishes to think about this in terms of phasality, this means that every head is in some sense a (PF) phase: the moment a head is merged, the complement will be linearized with respect to it. Whether this follows from phasality or from something else, is something we are agnostic about. The same holds for the workings of phasality or its alternatives on the LF side. The only thing that is crucial for is, is that linearization is highly cyclic.

that Koster 1974 famously observed. Concretely, Koster observed that in Dutch VPs with two specifier/adjuncts, only two of four possible linear orders are fine when they are both linearized at the same side of the verb:

- (33) a. dat Jan [[tijdens de pauze]₁ [[aan zijn vader]₂ dacht]].
 that John during the break of his father thought
 ‘that John thought of his father during the break.’
 b. ??dat Jan [[aan zijn vader]₂ [[tijdens de pauze]₁ dacht]].
 c. dat Jan [[dacht [aan zijn vader]₂] [tijdens de pauze]₁]
 d. ??dat Jan [[dacht [tijdens de pauze]₁] [aan zijn vader]₂]

The structural order is the one where the adjunct *tijdens de pauze* (‘during the break’) is structurally higher than the argument *aan zijn vader* (‘of his father’). That means that when the two phrases precede the head, the higher one should precede the lower one. However, when the two phrases follow the head, the higher one should follow the lower one.⁵

This pattern follows from our analysis. The moment the VP is linearized with respect to the higher head T (which we take to be head-initial, but nothing depends on that), T’s sister, the VP, must be linearized. For now, let’s ignore cases, where the adjuncts appear at different sides of the verb. Now, the first phrase the linearization algorithm targets is the higher adjunct (*tijdens de pauze*). That adjunct must then either appear to the immediate right of T, or at the very end of the phrase. The next phrase (the argument *aan zijn vader*) then must either follow the higher one, if that one is at linearized immediately after T, or appear to the immediate left of the adjunct if that one is linearized in phrase-final position. The argument *aan zijn vader* cannot appear in the position to the immediate right of T or in phrase-final position itself, as then it would contradict the earlier-established linearization statements concerning the adjunct.⁶

Of course, more needs to be said about the placement of heads in English. For one, lexical verbs are marked for tense and agreement, which we do not attribute to linear adjacency between V and T. Rather, we assume that tense morphology is not hosted on T but rather on V itself, which then Agrees with T (see also Bjorkman (2012) for proposals that Agree between T and V may be responsible for determining the way V is inflected). Evidence for that may also come from examples like:

- (34) Wolfgang played tennis on every Sunday. Von Stechow (2006)

The intended interpretation of (34) is one where past tense outscopes the distributive quantifier *every Sunday*, which in turn outscopes the lexical verb *play*, yielding the paraphrase in (35a). The scopal order where past tense would take scope at its surface position, i.e., under *every Sunday*, amounts to the reading in (35b), which is absent (Von Stechow 2003, 2005; Zeijlstra 2012).

5. Post-verbal adjuncts in English appear to behave differently — their order is apparently free, and the scope of these adjuncts does not seem to track linear order. See Takano (2003) for some discussion of approaches to these facts.

6. Note that when the adjuncts appear at different sides of the verbs, there is no issue, as both can appear in an edge position in either order.

- (35) a. There exists a past interval t such that for every Sunday in t , Wolfgang plays tennis.
 b. *For every Sunday, there exists a time before it such that Wolfgang plays tennis at that time.

This suggests that the semantic locus of tense does not coincide with the surface position of the tense marker. Hence, the tense marker must be realized in a position below T (and arguably on V).

The idea that tense is expressed on V is, of course, by no means new. In fact already the earliest proposals of Affix Hopping (Chomsky 1965) derive that, and in Distributed Morphology it is captured by means of M-merger (Marantz 1988; Halle and Marantz 1993). However, as neither Affix Hopping nor M-merger find a natural place in our proposal, we derive these facts by Agree and do not need to allude to additional (PF) operations for it.

Another thing that needs to be addressed concerns auxiliaries. As is well known, auxiliaries (*be*, *have*, modals) display certain head movement effects. Take, for instance, the following examples, where the auxiliary verb appears to the left of negation.

- (36) a. Mary hasn't left
 b. Mary can't leave

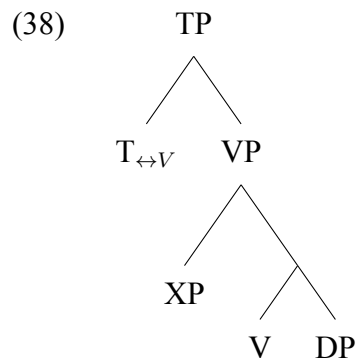
Have is clearly verbal and therefore must head a VP. Also, a modal like *can* takes scope below negation and must thus head a projection that is c-commanded by negation. Despite this, they appear to the left of the marker of negation. Their surface position, therefore, does not reflect their base position, which would commonly be attributed to head movement. We might then wonder how such head movement effects can result from our linearization algorithm. To answer this question, we will first focus on a language that manifests head movement effects for all verbs, namely French. After that, we will return to the question raised by English auxiliaries.

2.2 Adjacency diacritics

Now, let's look at French. As shown before, in French, the finite verb needs to precede adverbs that are located at the VP-boundary, unlike is the case with English lexical verbs. An adjunct thus follows the verb but may still precede or follow the object.

- (37) a. Jean embrasse souvent Marie
 J. kisses often M.
 b. Jean embrasse Marie souvent
 J. kisses M often.
 "John often kisses Mary."

Following classical analyses of head movement, we assume that in French T is subject to a condition that it is linearly adjacent to V. However, unlike classical analyses of head movement, we do not assume that this triggers head adjunction to T. We assume that in French T is equipped with a feature, which we notate as $[\leftrightarrow V]$, which merely informs the linearization mechanism to linearize V and T in a strictly adjacent fashion (see Ryan 2010). It ensures that the tense marker and the verb end up being adjacent so that the tense affix is not stranded. With this in mind, let us now look how linearization will take place exactly. Take the tree in (38).



First, after merger of V and DP, the following linearization construction is derived following the algorithm introduced earlier in this section, based on the fact that French is head-initial.

- (39) a. $V < DP$

Then, upon merger with T, an additional instruction is generated. Next to the instruction that says that T should precede VP, there is an additional instruction, $T \ll V$, which states that T should be strictly adjacent to V. We thus derive the following rules:⁷

- (40) a. $T < XP$
 b. $T \ll V$
 c. $T < DP$
 d. $V < DP$

The final question is then how XP must be linearized. What we know is that such elements have to be spelled out in the edge of T's complement, i.e. XP must be linearized as closely to T as possible with respect to other linearization conditions, or as far away as possible from A with respect to other linearization conditions. Since, V must be strictly adjacent to T, the closest possible linear position for XP to T is then to the immediate right of V. The position furthest away is immediately after DP. Thus, the following statements hold:

- (41) a. $T < XP$
 b. $T \ll V$
 c. $T < DP$
 d. $V < DP$
 e. $V < XP$
 f. $XP < DP$

or

7. Note that this may suggest that T should be prefixal on V instead of suffixal, contrary to fact. However, the internal order of V and T is something that is determined by the morphology/vocabulary insertion that applies after linearization

- (42) a. $T < XP$
 b. $T \ll V$
 c. $T < DP$
 d. $V < DP$
 e. $V < XP$
 f. $DP < XP$

Note that by the sole innovation of a diacritic like $\leftrightarrow V$ the linearization component can linearize the verb in a position that does not *appear* to reflect its base position. But crucially, no head movement has taken place, either in syntax or at PF. In French, T is uniformly equipped with $\leftrightarrow V$. One could even say that $\leftrightarrow V$ is connected to the feature [T]. What this derives is that T and V are linearly adjacent.

But this does not always have to be the case. It may very well be that only T with some particular feature has a property like $\leftrightarrow V$. This is what we assume is the case for English auxiliaries. We assume that $T[Aux]$ has a diacritic $\leftrightarrow V$, so that whenever T merges with a VP (or any other phrase) with a head containing feature [Aux], this head ends up strictly adjacent to T. The presence or absence of auxiliaries is then controlled by the presence or absence of [Aux] on T. If [Aux] is present, there must be an auxiliary in the clause, which then ends up adjacent to T. If no auxiliary is present, $\leftrightarrow V$ on T remains unsatisfied. Hence, only if [Aux] is absent on T, can verbs not raise to T. This is trivially the case when there is only a lexical verb inside VP. But what happens if there is an auxiliary inside VP without $T[Aux]$? One way we might rule this out is by appealing to selection — we could propose, for instance, that this would be blocked by the selectional properties of T, with $T[Aux]$ being incompatible with lexically headed VPs. Or it could be assumed that every auxiliary must stand in some Agree relation with $T[Aux]$.

Note that this also derives do-support. Since the negative marker *n't* can only be attached to auxiliaries, the sentence must contain an auxiliary. One can implement this by assuming that *n't* also is compatible with the selectional requirements of $T_{[Aux]}$.⁸ In addition the negative head must have a diacritic $\leftrightarrow V$. Then it follows that a sentence with sentential negation must contain an auxiliary verb, either *be*, *have*, a modal or *do*. In a sentence like *Mary doesn't leave*, *n't*, the complex unit $V-n't$ (where the negative marker suffixes on the verb), then further satisfies the adjacency requirement encoded by the $\leftrightarrow V$ diacritic on T as well.⁹

Naturally, the question arises why certain heads have adjacency features and others not. We assume that such diacritics reflect what underlies the Stray Affix Filter (Lasnik 1981). To avoid that some affix would be left without a head, an adjacency critic ensures that its host ends up at the right place. Note that this does not create a look-ahead effect. The fact that some form is morpho-phonologically bound forms a cue for the language learner that a particular head has some adjacency diacritic.

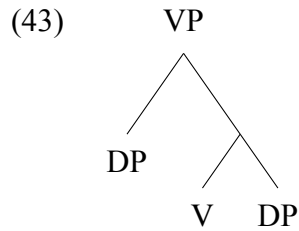
8. And in the case where the phrasal negative marker *not* is used, the negative head still carries such an agreement feature.

9. Strictly speaking, one could argue that in this case $\leftrightarrow V$ on T is not fully satisfied as *n't* still intervenes. However, since *n't* ends on the right of the auxiliary, this requirement is ultimately fulfilled upon Vocabulary Insertion. Alternatively, one could also argue that higher heads in the extended projection, such as the head of NegP, also carry a feature [V], and therefore count as verbal as well.

2.3 Solutions to previous problems

So far, we have established an account of head displacement effects that crucially involve no such thing as head movement. Heads do not undergo movement and stay at their base position when they are linearized. This mechanism naturally resolves the four problems that traditional accounts of head movement were facing: the *Cyclicity Problem*, the *Vacuity Problem*, the *Phrasing Problem*, and *Locality Problem*. As for the *Cyclicity Problem*, since there is no head movement going on whatsoever, this problem is trivially resolved. As for the *Vacuity Problem*, since linearization is a PF phenomenon, no semantic feeding is expected in the first place.

As for the *Phrasing Problem*, since heads do not undergo movement, the syntactic structure for a ditransitive should give rise to the prosodic structures discussed by Kalivoda (2018). Consider the following representation, alongside the assumption that V is to be linearized at the left edge of the VP, as a consequence of a relationship with some higher head.



Suppose we adopt an approach to the syntax-prosody interface like that proposed by (Selkirk 2011).

(44) **Match-XP**

For each XP, there is a phonological phrase ϕ dominating the exponents of the terminals XP dominates.

On such an approach, we are lead to expect the verb and both of its arguments to phrase together, as shown below. The VP maps to a ϕ , as do both of the arguments of the verb.

(45) $(V DP_{\phi} DP_{\phi})_{\phi}$

Crucially, we do not expect the phrasing shown below, which we would on a head-raising approach where V has moved to some higher position. There is no XP which dominates the two arguments to the exclusion of the verb, so such a phrasing will always violate Match-XP.

(46) $(V (DP_{\phi} DP_{\phi})_{\phi})_{\phi}$

The question that remains, then, is why the DPs themselves are not encapsulated in their own prosodic phrases. We can understand this as the result of a family of Binariness requirements on prosodic phrasing, which prefer prosodic phrases which consist of exactly two prosodic words (see Itô and Mester (1992) and Selkirk (2000) for more discussion of this point). Kalivoda's survey focuses on ditransitive constructions where the arguments of the ditransitive verb each consist of one prosodic word — while (47a) most closely satisfies Match-XP, it is in violation of Binariness, as the ϕ corresponding to each DP consist of a single phonological word each. (47b-c) come closer to respecting Binariness.

- (47) a. $(V DP_{\phi} DP_{\phi})_{\phi}$
 b. $(V DP DP)_{\phi}$
 c. $(V DP)_{\phi} (DP)_{\phi}$

It is useful here to spell out what the family of Binarity constraints might look like, following Selkirk (2011).

- (48) a. BinMin
 A ϕ should dominate no fewer than two other elements.
 b. BinMax
 A ϕ should dominate no more than two other elements.

We can see that both (47b-c), the most common patterns found by Kalivoda, as different strategies for satisfying the Binarity family above. (47b) violates BinMax while satisfying Match-XP and BinMin. (47c) violates BinMin while satisfying Match-XP and BinMax. The common patterns from Kalivoda's survey fall out as an interaction between Match-XP and the Binarity family. Crucially, the expected interactions only arise given an input structure where raising of V has not taken place.

Finally, as for the *Locality Problem*, the Head Movement Constraint follows as linearization is strictly local. To see this, assume that T has a diacritic $\leftrightarrow V$, but that v lacks any such diacritic. Then, specifiers and adjuncts of V can intervene in between V and v, and there is no way in which T could be guaranteed to end up adjacent to T. In fact, in a harmonic system, v will always appear in between T and v, hence T-V adjacency will never be available either. The only case where v would not appear in between T and V (and where T and V could end up adjacent to each other) are cases where T and V are head-final and v is head-initial, or the other way round. Both configurations are, however, independently ruled out, as they violate the Final-Over-Final Condition (FOFC, Biberauer, Holmberg, and Roberts (2014)). Hence, there is no way for T to end up adjacent to v, unless v has a $\leftrightarrow V$ and T a diacritic $\leftrightarrow v$. That derives the exact locality footprint of what looks like head movement.

2.4 So far

So far, we have established an account of head displacement effects that crucially involves no such thing as head movement. Heads do not undergo movement and stay at their base position when they are linearized. This mechanism naturally resolves the four problems that traditional accounts of head movement were facing: the *Cyclicity Problem*, the *Vacuity Problem*, the *Phrasing Problem*, and *Locality Problem*. As for the *Cyclicity Problem*, since there is no head movement going on whatsoever, this problem is trivially resolved. As for the *Vacuity Problem*, since linearization is a PF phenomenon, no semantic feeding is expected in the first place. As for the *Phrasing Problem*, since heads do not undergo movement, the syntactic structures should give rise to the relevant prosodic structures attested. And, finally, the *Locality Problem* also no longer applies, as the linearization mechanism is highly local itself; it applies at any instance of merger of a head with its complement.

We do so by making a number of assumptions, most notably that heads can be equipped with a diacritic that states how it is to be linearized with respect to its complement as well as whether it needs to be adjacent to some other head. These assumptions, as well as the ancillary assumptions we make,

all have been independently proposed and/or motivated in the literature before. As for the former, anybody who does not assume a specific linearization algorithm such as the LCA, must assume head complement linearization diacritics, and the adjacency diacritic is essentially an implementation of the Stray Affix filter. Hence, our overall theoretical apparatus is not in any way richer, if not poorer, than in alternative approaches to head movement.

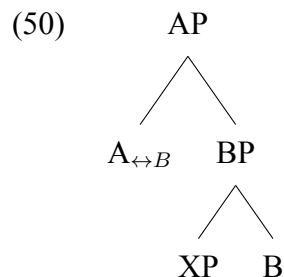
At the same time, there are also many questions left open. One of them concerns so-called "mixed headedness" patterns. These, we will discuss in the next section.

3 Mixed headedness

So far we have looked at harmonious languages where every head is either head-initial or head-final. In this section we discuss a potential challenge for the theory developed here — that of "mixed headedness". Those are languages where one head that is head-initial dominates another head that is head final. Take, for instance, German (49). German is a language with head-final VP and TP but with head-initial CP. In addition, in main clauses, the finite verb/auxiliary 'raises' to C. This way, the highest verb (either lexical or an auxiliary) shows up in a clause-initial position, preceding the object, but the rest of the verbs cluster in the hierarchically determined order in clause-final position (see Besten and Rutten 1989 and many others).

- (49) a. Ich sehe Marie.
 I see Marie
 "I see Mary."
 b. Ich habe Marie gesehen.
 I have Marie seen
 "I have seen Marie."
 c. Ich werde Marie gesehen haben.
 I will Marie seen have
 "I will have seen Marie."
 d. dass Ich Marie gesehen haben werde.
 that I seen Marie have will
 "that I will have seen Marie."

Apart from cases like (49d), the theory of linearization developed so far does not straightforwardly account for such cases where a head-final verb appears to 'raise' to C. The reason is that in such cases, the linearisation mechanism yields conflicting statements. To illustrate this, let's look at the tree below, including only a higher head-initial AP dominating a lower head-final BP, where B should be string-adjacent to A:



Here, B, being head-final, should be linearised to the right of its complement XP. At the same time, it should be strictly adjacent to A, and A should be linearised to the left of XP:

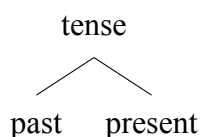
- (51)
- a. $A < B$
 - b. $A < XP$
 - c. $XP < B$
 - d. $A \ll B$

This is a contradictory set of linearization statements: (51d) precludes the possibility of the statements (51b) and (51c) in conjunction. In general, the system we have developed does not account for straightforwardly cases where one head may appear to the left or right of its complement in one context and on the other side in another. Below, we present a possible solution to this problem.

One approach we could consider involves a non-Kaynean version of an asymmetry between head-initiality and head-finality, inspired by markedness theory. In many cases, markedness effects are not taken to be the result of one of two features that stand in a geometrical sisterhood relation (see Harley 1994), one being more prominent than the other, but rather the result of the result of two features standing in a geometrical mother-daughterhood relation.

To see this, take present vs. past tense. One could argue that a geometry of tense features would involve a superfeature [Tense] with two daughters, [Past] and [Present]. Here, [Past] and [Present] would stand in a symmetric sisterhood relation and the alleged markedness of [Past] should follow from something else. Alternatively, one may assume that the geometry actually involves a mother feature [Tense] and a daughter [Past]. [Tense] is unspecified with respect to anteriority, [Past] is not. The fact that [Tense] generally denotes a present tense would then be a pragmatic blocking effect: anteriority must be expressed by using the more specific [Past] feature, hence only if the speaker does not intend to make an anteriority claim will she use underspecified [Tense]. Such an asymmetry between past and present has been proposed independently by many (including Sauerland 2002; Altshuler and Schwarzschild 2013 and many others). If [Past] and [Present] indeed stand in asymmetric feature-geometric relation, it follows naturally that the more specific feature is also the more marked one.

(52) **“Differentiated present” approach**

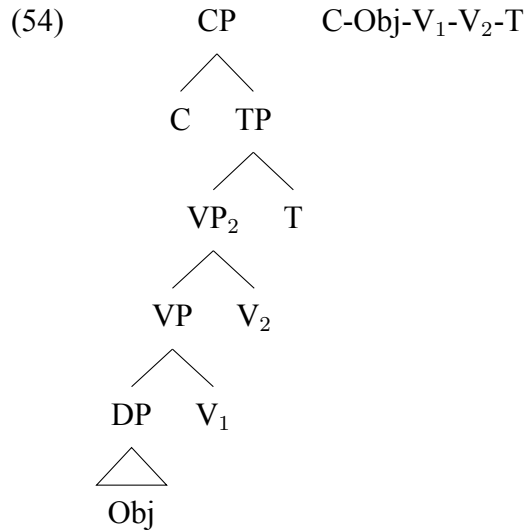


(53) **“Underspecified present” approach**



The same reasoning could apply to opposite featural diacritics like $<$ and $>$. They could be real duals of each other, with opposite head-directionality ‘values’, but they could also be taken to stand in an asymmetric relation. We could assume instead that there are two diacritics. One is $<$ (giving rise to head-initiality) and one is $\bullet$. $\bullet$ is underspecified. Since $<$ is more specific than $\bullet$, we take $\bullet$ to have a default interpretation of $>$ unless this violates other linearization statements (just like [tense] receives a non-past interpretation as long as that is compatible with the rest of the meaning). Hence, $H < \text{Comp}$ means H should precede its complement, $H \bullet \text{Comp}$ means that H follows Comp unless this violates other linearization statements.¹⁰

Now take German C-Obj-V-Aux, where Aux is to be treated as the morphological realization of an adjacent V and T.



Now assume C is head-initial and that V and T are head final. This gives rise to the following linearisation statements:

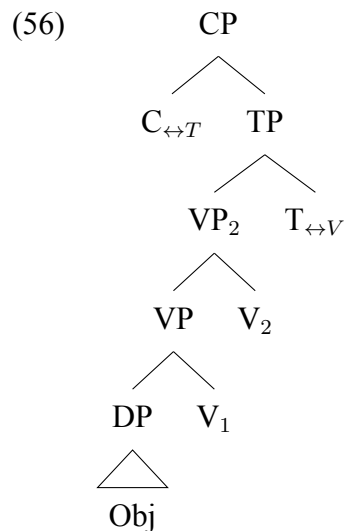
- (55)
- a. $C < TP$
 - b. $T \bullet VP_2$
 - c. $V_2 \bullet VP_1$
 - d. $V_1 \bullet \text{Obj}$

This means that T and V may strictly speaking appear either to the left or right of their complement, but unless anything contradicts this, TP and VP will be linearised in a head-final way.

Now, let's also assume the following linearisation conditions $T_{\leftrightarrow V}$ and $C_{\leftrightarrow T}$.

In that case, linearization of the following tree, where T should be adjacent to V ($T_{\leftrightarrow V}$) and C to T ($C_{\leftrightarrow T}$) would block interpretation of $\bullet$ as being a head-final diacritic.

10. A reviewer notes that this might motivate an Optimality Theoretic approach, rather than the particular algorithm for linearization we have specified here. We are relatively confident that an OT grammar could be written that would produce similar results to the algorithm we have utilized, and think that exploration of this idea may prove fruitful. However, we shy away from doing so in this paper — one expectation that we might have on an Optimality Theoretic approach is that learners should be able to rerank the constraints, giving rise to a multiplicity of potential linearization mechanisms. The expectation then would be that these mechanisms have empirical bite — that languages can be found where this clearly is the case. We are less confident that such a claim is correct.



To see this, let's go through all the relevant steps.

First step: $V_1 \bullet \text{Obj}$, will result in V ending up to the right of Obj, unless overruled later on:

- (57) a. $V_1 \bullet \text{Obj}$

Next, $V_2 \bullet \text{VP}_1$ will result in V_2 ending up the right of VP_1 , unless overruled later on.

- (58) a. $V_1 \bullet \text{Obj}$
 b. $V_2 \bullet V_1$
 c. $V_2 \bullet \text{Obj}$

In the third step, $T_{\leftrightarrow V} \bullet \text{VP}_2$, should in principle still result in T appearing to the right of VP_1 (unless overruled later on). The diacritic, will result in a statement that V_2 and T be adjacent. Hence, we have, where $\bullet\bullet$ indicates (unspecified) strict adjacency:

- (59) a. $V_1 \bullet \text{Obj}$
 b. $V_2 \bullet V_1$
 c. $V_2 \bullet \text{Obj}$
 d. $T \bullet\bullet V_2$
 e. $T \bullet V_1$
 f. $T \bullet \text{Obj}$

Here, nothing would block an interpretation of $\bullet$ as $>$, and the following order would appear: $\text{Obj} < V_1 < V_2 < T$. Note that this is indeed the attested order in German embedded (non-V2) clauses (49d).

But now, let's see what happens when matrix head-initial C with the adjacency diacritic for T, $\leftrightarrow T$, gets merged:

- (60)
- a. $V_1 \bullet \text{Obj}$
 - b. $V_2 \bullet V_1$
 - c. $V_2 \bullet \text{Obj}$
 - d. $T \bullet \bullet V_2$
 - e. $T \bullet V_1$
 - f. $T \bullet \text{Obj}$
 - g. $C \ll T$
 - h. $C < V_2$
 - i. $C < V_1$
 - j. $C < \text{Obj}$

As the reader can see, at this stage, in order not to trigger any contradictions, C must be linearized to the left of TP (and everything in it), while C must be strictly left-adjacent to T and T and V_2 must be strictly adjacent too. This will now force $\bullet$ to be interpreted as (non-default) $<$ in the following set of statements:

- (61)
- a. $V_1 \bullet \text{Obj}$
 - b. $V_2 < V_1$
 - c. $V_2 < \text{Obj}$
 - d. $T \ll V_2$
 - e. $T < V_1$
 - f. $T < \text{Obj}$
 - g. $C \ll T$
 - h. $C < V_2$
 - i. $C < V_1$
 - j. $C < \text{Obj}$

The only unspecified order is that between Obj and V_1 . Hence, only V_1 will be linearized to the right of the object. Hence, we get the following linearization: $C \ll T \ll V_2 < \text{Obj} < V_1$, which is the attested order. Thus, this way German V2 orders can be derived as well.

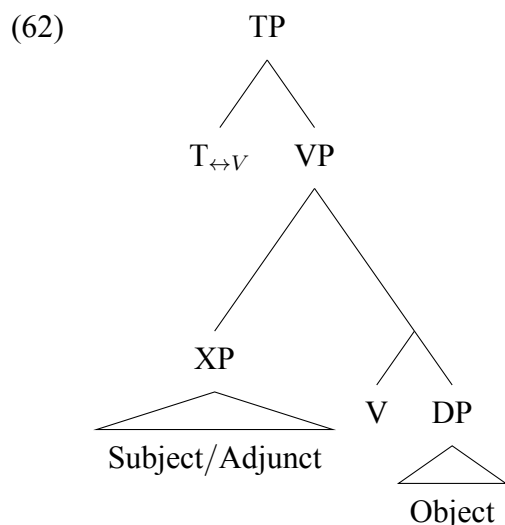
Note furthermore that the mirror image of these V2 orders (i.e. V-penultimate orders) cannot be derived given the asymmetry between head-finality and head-initiality. As long as it is hard-wired that head-initial features are more marked than head-final ones and markedness implies specificity relations, only leftward head 'movement' is possible. Of course, this begs the question as to why a mirror system would not be possible (i.e. $>$ and $\bullet$, where $\bullet$ would encode head-initiality when possible). The idea would be that in such a system rightward head movement effects but not leftward head movement effects should be attested. Whatever the exact nature is of the ban on rightward head movement effects (either in terms of filler-gap mechanisms or as part of inherent restrictions on grammar) is reflected in the availability of $<$ and $\bullet$ only.

4 Whether or not a language has scrambling

4.1 Scrambling and the linearization system

In this part of the paper we introduce another feature of heads, which requires them to be non-initial, in that they must host a (leftward) specifier of some sort. We suggest that this sort of feature underlies at least some classical EPP effects. That is to say: the need for an element to be non-initial might consistently be satisfied by syntactic movement of a phrase to a specifier position. In doing so, we highlight a potentially interesting property of the system developed here: elements that undergo what looks like head movement may not themselves be allowed to have an EPP property. Building on work by Branen (2021), we show that this derives a difference between Japanese and Tagalog with respect to the semantic effect of scrambling. In both languages — analogous to the cases discussed involving adjuncts in the prior section — complements of the verb and specifiers may be linearized freely with respect to each other, giving rise to semantically vacuous scrambling.¹¹

In the previous section, we assumed a simplified clause structure like that given below for French-like languages. There, we saw that complement and adjunct or specifier could be freely linearized with respect to each other, while V was required to precede both of them.



It turns out this model for French works very well for languages like Tagalog.¹² More specifically, a structure like that above is one where we should expect the verb to appear first, followed by the arguments it selects. Much like adverbs and objects in French, an agent in spec,VP and internal argument as comp,V will be freely ordered with respect to each other. As shown below, this is indeed the case.¹³

11. This is analogous, in some respects, to proposals made by Sauerland and Elbourne (2002) for scrambling in Japanese. There, they suggest that scrambling in Japanese may take place either in the narrow syntax or at PF. In the former case, reconstruction does not take place, while in the latter case the constituent is obligatorily interpreted in its base position. Considered in this light, the theory developed here builds a model of what PF scrambling might look like.

12. We set aside the question of voice alternations in Tagalog for now, and return to them at the end of this section.

13. We gloss *ang* here as a nominative marker, but note that everything said here is compatible with an analysis of ergative-absolutive analysis of Tagalog like that put forth in Aldridge (2004). We gloss *ng* as a genitive marker,

- (63) a. L<um>unon ang ina ng mani.
av.swallowed nom mother gen peanut
'The mother swallowed a peanut'.
b. L<um>unon ng mani ang ina.
av.swallowed gen peanut nom mother
'The mother swallowed a peanut'.

We should further expect that the relative order of arguments has no effect on scope or variable binding. As demonstrated below, this is indeed the case.¹⁴ Regardless of the order of a pivot agent and non-pivot theme, for instance, the theme may never take scope over the agent, and thus may not bind a variable in it.

- (64) a. *Nag-mamahal ng bawat anak ang kanyang ama
av-love gen every child nom poss. father
'Her_i father loves every_i child.'
b. *Nag-mamahal ang kanyang ama ng bawat anak
av-love nom poss. father gen every child
'Her_i father loves every_i child.'

Rackowski (2002)

This absence of semantic effect is just what we should expect if the free order of arguments in Tagalog is a result of the linearization algorithm developed here, rather than a result of syntactic movement (see, e.g. (Richards 1993) for an analysis of Tagalog scrambling as involving $\bar{A}$ -movement which undergoes obligatory reconstruction). However, given the fact that there are languages — such as Japanese, which we will discuss in further detail below — which do allow scrambling that has a semantic effect, we might now wonder why it would be the case that no learner of Tagalog takes the free order of constituents as a sort of syntactic scrambling. It turns out that the system we have developed has a ready answer for this question when amended to allow us to understand movement dependencies.

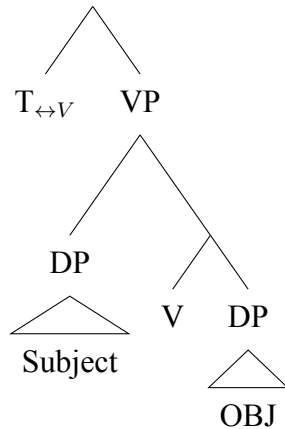
To illustrate how this works, we turn to Japanese, and will subsequently return to Tagalog. First, consider the two simple sentences like the following. We assume that Japanese is a head-final language — in the structures below T and V are linearized to the right of their complements. Note, given the logic we applied to the case of adjuncts in French, that the relative order of arguments is free. The subject may be linearized either to the left of V, but to the right of V's complement, thereby placing the subject as close as possible to the higher T, modulo the requirement that T and V be strictly adjacent, or to the left of both V and the complement of V. In the latter case, an SOV order arises, while in the former case an OSV order arises.

following standard convention where the *ng* marker which appears on many non-pivot arguments is analyzed as being identical to a comparable *ng* which appears on arguments of nominals.

14. See also Branen (2021) for discussion of nuclear stress in Tagalog that is in accord with what has been discussed here.

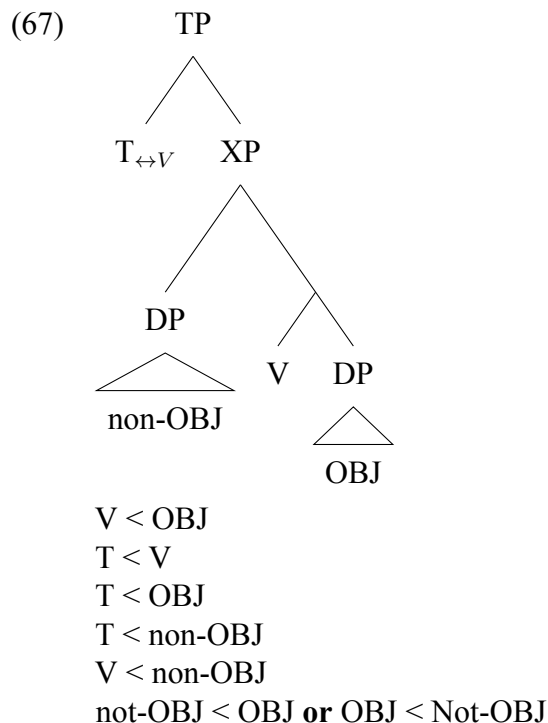
- (65) a. John-ga ringo-o tabe-ta
 J.-nom apple-acc eat-pst
 “John ate an apple.”
 b. Ringo-o John-ga tabe-ta
 apple-acc J.-nom eat-pst
 “John ate an apple.”

(66) TP OBJ < V



V < T
 OBJ < T
 SUBJ < V
 SUBJ < T
 SUBJ < OBJ **or** OBJ < SUBJ

Note that the underlying structure for the two sentences is the same, even in the case where the object precedes the subject. Given that Japanese has scrambling, this is somewhat desirable. For Tagalog, the facts are very much the same, modulo the setting of the headedness parameter. As sketched below, V will be linearized before the object, and the subject may be linearized between the verb and object — placing it as close to T as possible, modulo the requirement that nothing appear between V and T — or after both the verb and object — placing it as far away from T as possible. The effect here is much the same as in Japanese: the presence of an adjacency requirement between T and V results in an in-situ subject being linearized freely with respect to the object.

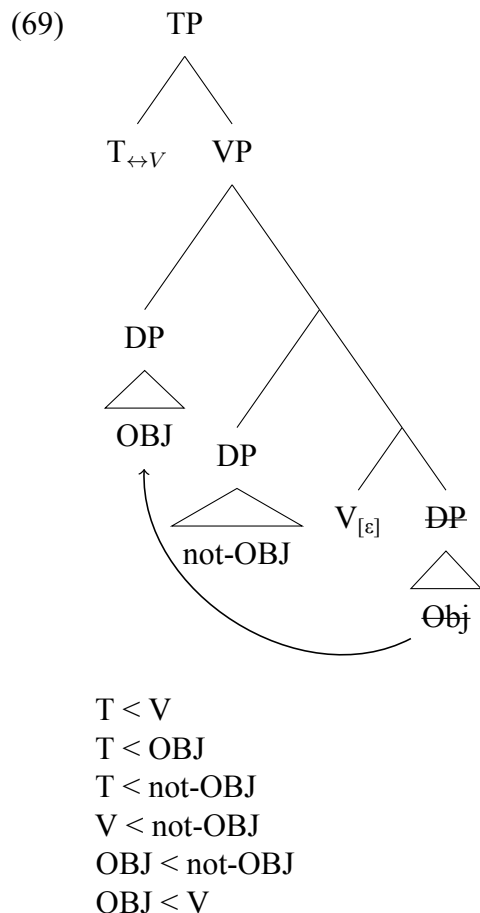


What of the cases where scrambling *does* have a semantic effect in Japanese? At present, our system is unable to account for these cases, and thus requires an amendment. As we have stressed throughout, our approach to linearization is *feature driven* — the basic insight is that the individual properties of the categories lexical items belong to are by and large the factors which determine the relative order of specifiers, heads, and their complements within a given maximal projection. Here, we would like to introduce another feature of lexical items, which we call $[\varepsilon]$ (where $[\varepsilon]$ is reminiscent of EPP, in that it is a feature which drives movement).¹⁵ For the system developed here, the order of specifiers with respect to other elements in their phrase is in principle free. We suggest that the $[\varepsilon]$ feature demands that a particular specifier of a phrase appear in the highest position. Features of this sort, we suggest, can also be triggers for movement of elements c-commanded by the head bearing the relevant feature: whichever element undergoes movement driven by $[\varepsilon]$ must be leftmost in the relevant phrase.¹⁶

For the cases of scrambling in Japanese which do have a semantic effect, we assume a structure like the following. Here the object has undergone movement to a specifier position, driven by the needs of $[\varepsilon]$. Consequently, the object must be linearized at the left edge of VP, allowing only the OSV word order. Here, the object precedes the subject, feeding an interpretation where the object takes scope over the subject for the purposes of evaluating scope and variable binding.

15. Essentially, we adopt a feature-driven approach to scrambling. See Grewendorf and Sabel (1999), and Erlewine and Lim (2023) for some recent discussion of locality conditions on scrambling on such an approach

16. Alternatively, one could assume that there is a higher head present that triggers movement of the relevant phrase into its specifier. This assumption is orthogonal to our proposal.



The system presented here should allow Japanese-like scrambling only in languages that are head-final, or head-initial without adjacency requirements between V and T like those found in French. In languages that do have such an adjacency requirement, scrambling will be impossible, as the particular linear order imposed by $[\varepsilon]$ is incompatible with the linear order imposed by $[\leftrightarrow V]$ on T.¹⁸

4.2 Voice alternations in Tagalog

Tagalog is famous for having a robust *voice system*, where one argument receives a particular marker which corresponds with systematic changes in the morphology of its selecting predicate. Much work on Tagalog voice alternations suggests that this privileged element — which we will call the *pivot* — occupies a structurally privileged position above all other arguments in the clause. For instance, for the *patient voice* clause below, there is good evidence that the pivot has undergone movement to a position above the agent: the theme pivot may bind a variable in the non-pivot agent.

18. There is at least some reason to believe this might be on the right track. Otsuka (2005) argues that Tongan (a V-initial language) has scrambling with a semantic effect; Tongan seems to lack an adjacency requirement between V and T, as clitic pronouns may appear in such a position. At least some well-studied head-final languages — such as Korean, Mongolian, and Turkish — behave like Japanese with respect to scrambling. An exhaustive review of this would be well beyond the scope of this paper, but there is reason to believe this proposal may be borne out to some extent.

- (70) M<in>amahal ng kanyang ama ang bawat anak
 pv.love gen poss father ang every child
 “Her father_i loves every child_i.”

Rackowski (2002)

Evidence for movement of this sort comes from facts involving scope and binding, where theme arguments may outscope and bind into agents only when they are the pivot of the clause (Rackowski 2002), as well as a process of quantifier float which may take place only from the pivot of the clause (Kroeger 1993).

This potentially seems at odds with the proposal put forth in this section, depending on the position one assumes non-agent pivots land in. If we adopt a more elaborated clause structure, we might expect movement of this sort to target a position “in the middle of the clause” — which should be ruled out given the proposal put forth here, which leads us to expect movement of this sort to uniformly be banned. Here, we present two possible ways out of this conundrum.

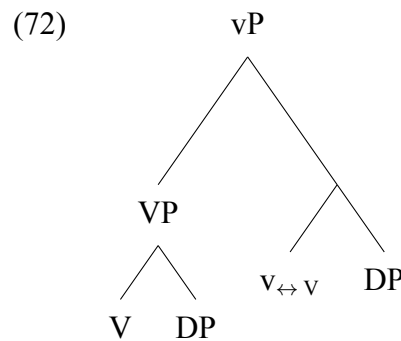
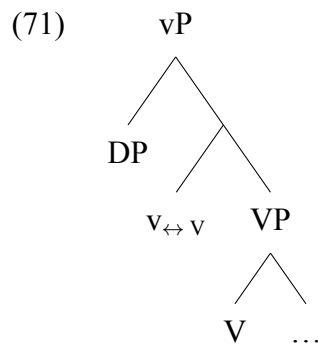
The first would be to deny that the landing site of the pivot is clause-medial in Tagalog. More specifically, we would hope that the pivot lands in a position at or above the highest head in the extended inflectional projection which bears a $\leftrightarrow$ feature. Such proposals have antecedents in the literature — McGinn (1988), Richards (1996), and Sabbagh (2005) all propose that the landing site of the pivot in Tagalog is somewhere high in the left periphery, but differ in their details as to how the pivot appears freely in the post-verbal position. McGinn and Sabbagh both suggest that pivots move to this position, and are then positioned freely after the verb through a post-syntactic operation. Richards suggests that movement to the pivot position in Tagalog is covert movement. Of the three, Richards’ proposal is the most straightforwardly compatible with the analysis that we have developed so far. We leave serious consideration of reconciling McGinn’s and Sabbagh’s proposals with our own a task for future work.

The second would be to deny that the pivot undergoes movement. Such an approach has been posited by Y.-H. Chen (2017). Chen suggests that Agree underlies the relationship between the lexical elements that are realized as voice morphology and the pivot, but that this does not involve displacement of pivots. This approach would be straightforwardly compatible with the theory developed here, with the choice of pivot having no effect at all on the syntactic configuration of arguments with respect to each other. To our mind, the main challenge such an approach faces is not in reconciliation with the theory developed here, but rather in explaining the aforementioned scope and binding facts in a parsimonious way. We leave this too a task for future work.

5 Conclusion

In this paper, we have set out to model classical head movement effects as the consequence of a particular linearization mechanism, rather than a structure-altering transformation. On this approach, classical head movement effects involve linearizing a head at the edge of the phrase it heads, with other orders being blocked as a consequence of the PF needs of the next head up. We have demonstrated how this approach works to derive the difference between French-like and English-like languages. We discussed two potential extensions of the model to account for mixed-headed languages like German. Finally, we saw that the theory leads us to expect scrambling to be possible in some languages but impossible in others.

An open question remains: why do adjacency requirements of this sort exist, if they appear to duplicate the already extant selectional relationship between a selecting head and its complement? One possible reason relates to recent proposals about phrase structure developed by Newman (2021). Newman argues at length that selectional requirements cannot guarantee that a selected element will necessarily be a complement or specifier, and that if that is the case, a variety of facts — such as apparent phrase structure paradoxes — receive a straightforward resolution. If this is on the right track, it is not clear that the duplication between complementation and adjacency is in fact duplication. By way of example, she proposes that a *v* which selects an external argument and internal argument should allow both phrase structures given below.



Such structures pose a challenge for classical approaches to head movement (as Newman discusses): in a language that would be argued to have movement of *V* to *v*, it is not entirely straightforward how head movement is supposed to take place, given standard assumptions about head movement. One advantage of the theory developed here is that it is able to straightforwardly ensure that *V* and *v* are adjacent: for both structures, *v* may be linearized in a position adjacent to the *V* in *VP*. In (71), this comes about straightforwardly; in (72) this could come about when *VP* is linearized with respect to *v*: *v* would have to be linearized to the left of *VP*, allowing *v* and *V* to be adjacent.

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